

FATEMEH MIRZAABOLHASSANI

Computer Science

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FIELDS OF INTEREST

- Tumor-Immune Microenvironment Analysis
- Single-Cell & Spatial Transcriptomics
- Gene Regulatory Network Inference
- Cancer Cell-State Transitions & Tumor Heterogeneity
- Cancer Genomics & Clonal Evolution
- Multi-Omics Integration for Precision Oncology

EDUCATION

Master of Computer Science

-  Oct. 2020 – Sept. 2023
- GPA: 17.80/20 with total 32 passed credits
-  Sharif University, Tehran, Iran
- Class Rank: Third

Bachelor of Computer Science

-  Sept. 2016 – Sept. 2020
- GPA: 19.44/20 with total 136 passed credits
-  Alzahra University, Tehran, Iran
- Class Rank: First

Mathematics and Physics High School Diploma

-  Sept. 2012 – Aug. 2016
- GPA: 18.98/20
-  Farzanegan High School, Tehran, Iran

SKILLS

Programming & Scripting: Python, R, Bash, Git/GitHub.

Single-cell & Bioinformatics tools: Seurat, Monocle3, SoupX, Harmony, pySCENIC, CellOracle, TradeSeq, SCPA, InferCNV, CopyKAT, Phylogenetics, Galaxy, Cytoscape, BioRender.

Molecular modeling & MD: GROMACS, Amber, AutoDock Vina, trajectory analysis, PyMOL.

Deep Learning & Machine learning & Statistics: Machine learning pipelines, statistical testing, scikit-learn, PyTorch.

HPC & Reproducibility: Docker, conda, renv, pipeline development.

MANUSCRIPTS IN PREPARATION

- Mirzaabolhassani F., et al. "Integrative Single-Cell Transcriptomics to Identify Regulatory Drivers in the Dedifferentiation of Thyroid Cancer." *Co-first author. Manuscript in preparation.*

RESEARCH EXPERIENCE

Integrative Single-Cell Transcriptomics to Identify Regulatory Drivers in the Dedifferentiation of Thyroid Cancer

-  Sep. 2024 – present
- **Role:** Lead author for bioinformatics analyses (co-first contribution; overall second author).
- Integrated trajectory inference, GRN analysis (pySCENIC, CellOracle), and CNV inference (CopyKAT, InferCNV) to identify candidate transcription factors and genes driving the **Normal → PTC → ATC** transition; results under experimental validation with clinical collaborators at EMRI, Tehran University of Medical Sciences.
- Supervisors: Dr. Kaveh Kavousi and Dr. Vahid Haghpanah

Identification of Driver Genes in Glioblastoma Based on Single-Cell Gene Expression Data Utilizing Pseudotime and Phylogenetic Analysis

📅 Oct. 2022 – present

- Developed a scoring framework integrating pseudotime-ordered expression dynamics with evolutionary distances inferred from mutation data, enabling prioritization of genes whose transcriptional changes are phylogenetically consistent with driver-level selection in GBM progression.
- This work formed the basis of my M.Sc. thesis and is currently being extended with additional analyses.
- Supervisors: Dr. Mohammad Hadi Foroughmand Araabi, Dr. Kaveh Kavousi, Dr. Fatemeh Zare-Mirakabad

Molecular Dynamics Simulation of Doxorubicin Binding to DNA and Topoisomerase II

📅 Jun. 2025 – present

- Building an MD workflow to simulate doxorubicin–DNA binding and stabilization of specific binding poses, in collaboration with biologists and computational physicists.
- Supervisors: Dr. Arash Boochani and Dr. Fatemeh Zare-Mirakabad

Implementation of RNA Secondary Structure Prediction

📅 Feb. 2024 – May 2024

- Implemented methods from “RNA Secondary Structure Prediction Using Stochastic Context-Free Grammars and Evolutionary History”; supervised a B.Sc. student. 🔗 [GitHub repository](#)
- Supervisor: Dr. Fatemeh Zare-Mirakabad

PRESENTATIONS

Identification of Driver Genes in Glioblastoma Using Single-Cell Gene Expression, Pseudotime, and Phylogenetic Analysis

- Poster presentation: Feb. 2025, Iran Conference on Bioinformatics, Zanjan, Iran. 🔗 [Abstract \(P.140\)](#) 🔗 [Poster](#)
- Oral presentation: Apr. 2025, Computational Biology Research Center, Amirkabir University, Tehran, Iran.

WORK EXPERIENCE

Bioinformatics Research Collaborator

📅 Sep. 2024 – present

📍 EMRI/TUMS, Shariati Hospital, Tehran, Iran

- Provide computational analyses for a clinical scRNA-seq study on thyroid cancer progression, working directly with clinicians and experimental biologists to translate computational findings into experimentally testable hypotheses.
- Collaboration with the Endocrinology and Metabolism Research Institute, Tehran University of Medical Sciences.

Research Affiliations

📅 2022 – present

- Computational Biology Research Center (CBRC), Amirkabir University of Technology (Polytechnic)
- Complex Biological Systems and Bioinformatics (CBB), University of Tehran

Workshop Instructor

📅 Sep. 2024

📍 Institute Pasteur, Tehran, Iran

- Conducted training on Next-Generation Sequencing techniques.

Teaching Assistant

📅 2018 – 2023

📍 Sharif University, Amirkabir University, Alzahra University

- Teaching assistant for Foundations of Computation Theory, Database, and Data Structures and Algorithms.

LANGUAGES

English

- IELTS: 7 (2022 – 2024), Next exam expected in July 2026

HONORS🏆

- M.Sc. Fellowship Award at Amirkabir University of Technology, Tehran University, and Sharif University as an exceptionally talented student (exempt from entrance exam).
- Ranked first in my B.Sc. class for three consecutive years.
- Received governmental fellowships for outstanding Bachelor performance.
- Admitted to NODET (National Organization for Development of Exceptional Talent) via competitive entrance exam for high school.

LEADERSHIP & EXTRACURRICULARS

Coach, ICPC (International Collegiate Programming Contest, Asia Region, Tehran site)

📅 2025

- Mentored and guided a competitive programming team, coordinating strategy, training, and problem-solving.

Co-founder, Computer Science Journal Club

📅 Winter 2018 – Summer 2020

REFERENCES

Dr. Kaveh Kavousi

Associate Professor, Dept. of Bioinformatics, Institute of Biochemistry & Biophysics, University of Tehran
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Dr. Fatemeh Zare-Mirakabad

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Dr. Vahid Haghpanah

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